

Numerical Methods [CH-2-16 (MO)]

End Semester Examination

Maximum Marks: 70

Maximum Duration: 3 hours

Scientific calculators are permitted

1. Determine with reasoning if the given PDE is parabolic, hyperbolic or elliptic.

$$\frac{\partial^2 V}{\partial t^2} = \frac{1}{2} \eta \frac{\partial^2 V}{\partial x^2} - r x \frac{\partial V}{\partial x} + r V \quad (2 \text{ Marks})$$

2. What type of PDE are each of the following? (1 Mark)

(a) Heat/Diffusion equation (b) Laplace/Poisson equation (c) Wave equation
~~parabolic~~ ~~elliptic~~ ~~hyperbolic~~

3. Write down the iterative scheme for the wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ using the central difference approach. If $c=1$, $\Delta x=0.1$, and $\Delta t=0.05$ compute the value u_5^2 given $u_4^1=0.2$, $u_5^1=0.4$, $u_6^1=0.2$, $u_5^0=0.3$. (5 Marks)

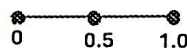
4. The 2-D heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ is defined on a grid $0 \leq x \leq 1$ and $0 \leq y \leq 0.5$ with $h=0.25$. Using Dirichlet conditions $u(x, y, t)=0$ on all boundaries, and initial conditions $u(x, y, 0) = \sin(\pi x) \sin(\pi y)$ at all interior points, compute 2 iterations of the solution using Gauss-Seidel update. (7 Marks)

5. Perform 2 iterations of Newton's method to minimize the given function $f(x) = 3 + 6x + 5x^2 + 3x^3 + 4x^4$ using the below central difference approximations $f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$ and $f''(x) = \frac{f(x+\Delta x) + f(x-\Delta x) - 2f(x)}{(\Delta x)^2}$. Use $\Delta x=0.01$ and initial value $x_0 = -1$ (7 Marks)

6. Compute 2 iterations of gradient descent update which minimizes the function $f(x) = x^4 - 3x^3 + 2\sin(2x) + 5x$. Use $\lambda=0.01$ and $x_0 = 1$ (3 Marks)

7. Evaluate the gradient vector and Hessian matrix of the equation $f(x, y) = 2xy^2 + 3e^{-y}$ at the point $(0,0)$. (3 Marks)

8. Show that Simpson's integration of the function $\int_0^1 x^2 dx$ on the following grid achieves the exact value. Explain why this is the case. (5 Marks)



9. In the method of Gaussian quadrature for integration, find the degree of the Legendre polynomial required to evaluate $\int_{-1}^1 (5x^4 + 3x^3 - 2x) dx$ (2 Marks)

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10. Derive the expression $f'(x) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$ using the Taylor series expansion and show that this has the accuracy of order 2 [$O(h^2)$] (5 Marks)

11. From the table below, find $f'(0.6)$ using (a) forward difference (b) backward difference (c) central difference. (3 Marks)

x	0.5	0.6	0.7
f(x)	3.5	4.25	3.9

12. Given the points $(x, y) = (-2, 5), (1, -7), (3, 4)$ construct the Lagrange basis polynomials (you need not simplify them) (4 Marks)

13. Given the following Newton's divided difference table, construct the polynomial and use the expression to analytically find the first derivative of the function at $x = 0.7$. (6 Marks)

	x	f(x)	Δ_1	Δ_2
x_0	0	0	0.958	-0.2346
x_1	0.5	0.4794		
x_2	1.0	0.8415	0.724	

14. Find the L matrix corresponding to the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ by performing row echelon operations. (4 Marks)

15. For the system of equations, find the Jacobian matrix evaluated at the point (1,1,1).

$$x^2 + y^2 + z^2 = 0$$

$$xy + yz + zx = 4$$

$$x + y - z = 1$$

(6 Marks)

16. Consider the function $f(x) = x^3 - x - 2$. Show that the function has a root in the interval [1,2]. Perform 2 iterations of the bisection method starting with this interval to approximate the root. (3 Marks)

17. Using the expansion $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ find the percentage truncation error of $e^{0.5}$ if the series is limited to (a) first term (b) first 2 terms (c) first 3 terms. (4 Marks)

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Please note the following changes in the question paper :

4.	The 2-D Laplace equation $\nabla^2 u = 0$ is defined on a grid $0 \leq x \leq 1$ and $0 \leq y \leq 1$ with $h = 0.25$ Using the Dirichlet condition $u(x, y) = 0$ on all boundary points, compute 2 iterations of the solution using Gauss-Seidel update. Use $u(x, y) = \sin(\pi x) \sin(\pi y)$ as the initial guess (7 Marks)
13.	Please read the columns as $x, f(x), \nabla_1, \nabla_2$